Do Now · retrieval

Karnaugh maps 1: start with four squares

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Write −13 as an 8-bit two’s-complement integer .

Q2

For the circuit shown, simplify Q = ¬(A ∨ B) using De Morgan’s Law.

Q3

Complete one full left-to-right pass of ascending bubble sort. What is the resulting array?

Q4

Normalise this positive binary mantissa without changing its value. Give the new exponent.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Map the truth-table outputs — For each combination of A and B, find the same output Q in the map. The grid moves outputs; it does not change them.

Core idea 2

Group adjacent 1s — Circle the two 1s in the bottom row. A group can contain 1, 2, 4, 8... adjacent 1s. Use the largest useful group.

Core idea 3

Compare the input values — For the two circled cells: — CODE:
A = 1, B = 0
A = 1, B = 1

Key visual · complete concept

A rows	BC columns 00	01	11	10
0	0	1	1	0
1	0	1	1	0

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Why Gray code? — From 01 to 10 , two bits change — these cannot be neighbouring K-map columns. — Each neighbouring pair differs by one bit only , including the end pair 10 ↔ 00 .

Core idea 5

Read the three-input layout — Two bits label each BC column. Moving sideways changes one input only. This allows useful groups in three-variable maps.

Vocabulary · use the correct technical terms

Karnaugh map — A grid used to simplify the outputs from a truth table.

Group — Adjacent 1s circled as one rectangular region.

Adjacent — Cells next to each other horizontally or vertically.

Simplify — Express the same output with fewer Boolean variables.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

How does this four-square map simplify?

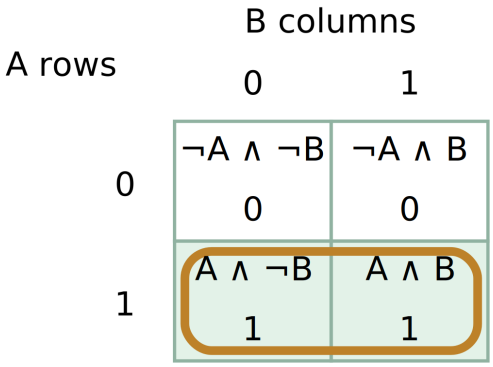
Follow the worked method

1. Read the input combinations — Only the two bottom cells contain 1. They represent A=1, B=0 and A=1, B=1 .

2. Compare each input

3. Write the simplified expression — Keep A because it remains 1. Remove B because it changes between 0 and 1. — Check: Q is 1 for both values of B whenever A = 1.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

1. Identify the group
Circle the two adjacent 1s.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

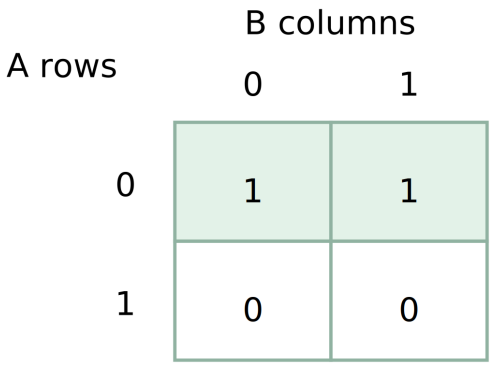
Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

One horizontal pair

Circle both adjacent 1s. For A and B, state which stays constant or changes. Write the simplified Q.

Question 1 · original source diagram



Check that you have used the technical terms and explained your decisions.

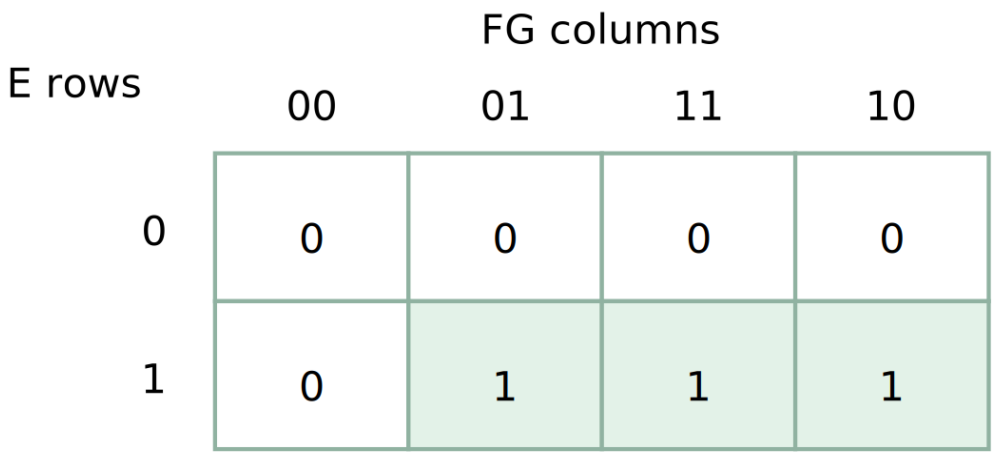
You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Free-popcorn voucher
E = voucher shown, F = soft drink, G = water. Free popcorn is offered when E AND (F OR G). Circle two useful pairs and write R as an OR of two AND terms. Adapted from OCR Boolean Algebra Q2(b).

Question 2 · original source diagram



Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A variable that changes value inside a Karnaugh-map group is removed from that group's simplified term.

Use key vocabulary from memory

1. Explain Karnaugh map in your own words.

2. Explain Group in your own words.

3. Explain Adjacent in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.